

HOMEWORK 9
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

SECTION 3.2 - POWER SERIES AND TAYLOR'S THEOREM

1. Compute the Taylor series of the following:

(a) $\frac{\sin z}{z}$, $z_0 = 1$

(b) $z^2 e^z$, $z_0 = 0$

(c) $e^z \sin z$, $z_0 = 0$

2. Compute the first few terms in the Taylor expansion of $\sqrt{z^2 - 1}$ around 0.

3. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converge for $|z| < R$. If $0 < r < R$, show that $f(z) = \sum_{n=0}^{\infty} a_n r^n e^{in\theta}$, where $z = re^{i\theta}$ and

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(re^{i\theta}) e^{-in\theta} d\theta.$$

Also show that

$$\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.$$

Hint: Use the Cauchy Integral Formula for a_n and expand $f\bar{f}$ in a series. Then, integrate term by term.

4. Find a function such that $f(0) = 1$ and $f'(x) = x + 2f(x)$ for all x .

SECTION 3.3 - LAURENT SERIES AND THE CLASSIFICATION OF SINGULARITIES

5. Find the Laurent series expansion of $\frac{z}{z+1}$ around $z_0 = 0$ valid in the region $1 < |z| < \infty$.

6. Prove, using the Taylor series, the following complex version of l'Hopital's rule:

Let $f(z)$ and $g(z)$ be analytic, both having zeros of order k at z_0 . then, $f(z)/g(z)$ has a removable singularity and

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f^{(k)}(z_0)}{g^{(k)}(z_0)}.$$